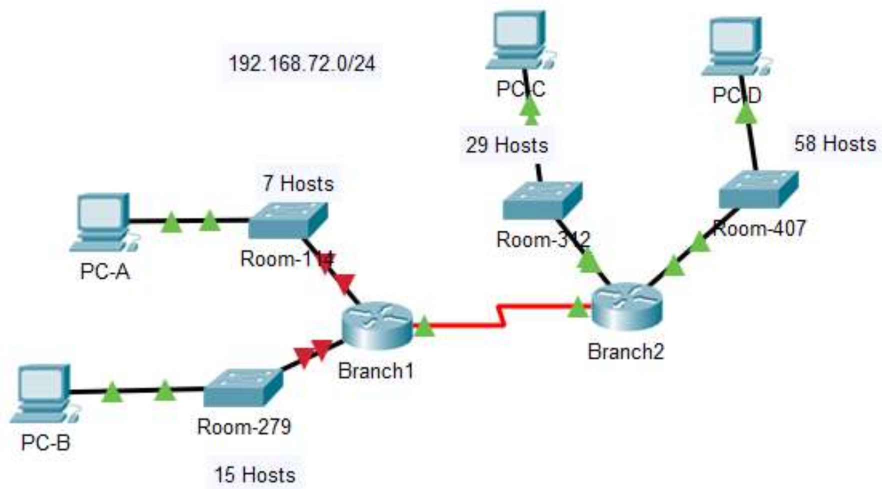
Lab6- Designing and Implementing a VLSM Addressing Scheme

Chapter 8 Activity

Topology



Resources

Packet Tracer

File needed: Lab6-VLSM.pka

Objectives

Part 1: Examine the Network Requirements

Part 2: Design the VLSM Addressing Scheme

Part 3: Assign IP Addresses to Devices and Verify Connectivity

Background

In this activity, you are given a /24 network address to use to design a VLSM addressing scheme. Based on a set of requirements, you will assign subnets and addressing, configure devices and verify connectivity.

Part 1: Examine the Network Requirements

Step 1: Determine the number of subnets needed.

You will subnet the network address 192.168.72.0/24. The network has the following requirements:

- **Room-114** LAN will require **7** host IP addresses
- **Room-279** LAN will require **15** host IP addresses
- **Room-312** LAN will require **29** host IP addresses
- **Room-407** LAN will require **58** host IP addresses

How many subnets are needed in the network topology?

Step 2: Determine the subnet mask information for each subnet.

a. Which subnet mask will accommodate the number of IP addresses required for **Room-114**?

   How many usable host addresses will this subnet support?

b. Which subnet mask will accommodate the number of IP addresses required for **Room-279**?

   How many usable host addresses will this subnet support?

c. Which subnet mask will accommodate the number of IP addresses required for **Room-312**?

   How many usable host addresses will this subnet support?

d. Which subnet mask will accommodate the number of IP addresses required for **Room-407**?

   How many usable host addresses will this subnet support?

e. Which subnet mask will accommodate the number of IP addresses required for the connection between **Branch1** and **Branch2**?

Part 2: Design the VLSM Addressing Scheme

Step 1: Divide the 192.168.72.0/24 network based on the number of hosts per subnet.

a. Use the first subnet to accommodate the largest LAN.

b. Use the second subnet to accommodate the second largest LAN.

c. Use the third subnet to accommodate the third largest LAN.

d. Use the fourth subnet to accommodate the fourth largest LAN.

e. Use the fifth subnet to accommodate the connection between **Branch1** and **Branch2**.

Step 2: Document the VLSM subnets.

Complete the **Subnet Table**, listing the subnet descriptions (e.g. Room-114 LAN), number of hosts needed, then network address for the subnet, the first usable host address, and the broadcast address. Repeat until all addresses are listed.

Subnet Description	Number of Hosts Needed	Subnet address	Network Address/CIDR	First Usable Host Address	Broadcast Address
0					
1					
2					
3					
4					

Step 3: Document the addressing scheme.

a. Assign the first usable IP addresses to **Branch1** for the two LAN links and the WAN link.

b. Assign the first usable IP addresses to **Branch2** for the two LANs links. Assign the last usable IP address for the WAN link.

c. Assign the second usable IP addresses to the switches.

d. Assign the last usable IP addresses to the hosts.

Part 3: Assign IP Addresses to Devices and Verify Connectivity

Most of the IP addressing is already configured on this network. Implement the following steps to complete the addressing configuration.

Step 1: Configure IP addressing on Branch1 LAN interfaces.

Step 2: Configure IP addressing on Room-312, including the default gateway.

Step 3: Configure IP addressing on PC-D, including the default gateway.

Step 4: Verify connectivity.

You can only verify connectivity from Branch1, Room-312, and PC-D. However, you should be able to ping every IP address listed in the **Addressing Table**.

Fill in the following table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Br1	G0/0			N/A
	G0/1			N/A
	S0/0/0			N/A
Br2	G0/0			N/A
	G0/1			N/A
	S0/0/0			N/A
Room-114	VLAN 1			
Room-279	VLAN 1			
Room-312	VLAN 1			
Room-407	VLAN 1			
PCA	NIC			
PCB	NIC			
PCC	NIC			
PCD	NIC			

